

Prudhvi Nelavelli

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Summary

Software Development Engineer with strong background in designing and optimizing backend web applications and APIs using C# and ASP.NET Core. Proven track record improving system performance-cutting API response times by up to 85% through caching, query tuning, and secure JWT-based authentication. Skilled in SQL Server, microservice architectures, and Azure, with collaborative approach to delivering scalable, reliable solutions.

Core Skills

- **Languages:** C#, JavaScript, TypeScript, SQL, Python, Java, HTML, CSS
- **Backend (.NET):** ASP.NET Core, Web API, MVC, Entity Framework Core, LINQ, Identity, Razor Pages, Middleware, Dependency Injection
- **Frontend:** Angular, React, RxJS, TypeScript, Bootstrap, Material UI, Responsive UI
- **Databases:** SQL Server, Azure SQL, PostgreSQL, MySQL, MongoDB, Redis
- **Cloud:** Azure (App Services, Functions, Key Vault, SQL, Blob Storage), AWS (Lambda, S3, RDS, ECS), Docker, Kubernetes
- **Architecture:** REST APIs, Microservices, CQRS, Clean Architecture, DDD, Caching, API Gateways, JWT OAuth2 Auth
- **Tools:** Visual Studio, VS Code, Postman, Swagger, Git, Azure DevOps, GitHub Actions, Jenkins
- **Testing:** xUnit, NUnit, MSTest, Moq, Integration Testing
- **Messaging:** RabbitMQ, Azure Service Bus, AWS SQS
- **Observability:** Azure AppInsights, Serilog, ELK, CloudWatch, Logging Pipelines
- **Methodologies & Concepts:** Computer Science Concepts, Software Development Lifecycle, Object-Oriented Programming, Agile Development

Certifications

- Microsoft Certified: Azure Developer Associate
- HackerRank: Problem Solving (Basic)
- HackerRank: Problem Solving (Intermediate)

Experience

Fiserv

Aug 2024 - Present

Software Development Engineer (Contract)

Columbus, OH

- Built and enhanced ASP.NET Core services for financial document processing and billing workflows, adhering to software development lifecycle best practices which increased system reliability and maintained performance during batch and peak loads
- Designed REST APIs with clear contracts using controllers, DTOs, validation rules, and middleware, leveraging object-oriented programming principles to deliver consistent error responses, improve request tracing, and strengthen authorization boundaries
- Implemented secure authentication and authorization patterns using JWT and role-based access, which protected customer data and ensured least-privilege access for internal consumers
- Improved data access performance by refactoring EF Core queries and applying algorithmic optimization based on computer science concepts, introducing compiled queries for hot paths, and collaborating on SQL indexing strategies for high volume tables
- Introduced Redis caching for frequently accessed metadata and reference lookups, reducing repeated database calls and stabilizing response times for read heavy endpoints.
- Implemented asynchronous processing using RabbitMQ and AWS SQS to offload long-running tasks, reduce synchronous bottlenecks, and improve overall throughput of downstream services
- Containerized services with Docker and deployed them to AWS ECS and EKS, validating configuration via environment variables, secrets, and health checks, which ensured reliable releases and reduced deployment errors
- Strengthened observability with Azure AppInsights, Serilog structured logs, distributed tracing, and dashboards to support quicker root cause analysis during incidents.
- Partnered with engineering and product stakeholders in design reviews to define service boundaries, API standards, and integration expectations for new modules, following agile development methodologies which aligned teams and accelerated development timelines

ComplexCare Solutions

Jun 2024 - Aug 2024

Software Engineer Intern

Bowie, MD

- Delivered healthcare features using ASP.NET Core Web API, EF Core, SQL Server, and Angular, enabling clinicians to complete workflows more efficiently and reducing manual data entry
- Built reusable Angular components for tables and forms with sorting, filtering, pagination, and responsive layouts, ensuring consistent UI across modules and decreasing development time for new screens
- Implemented API layers following clean architecture principles, separating controllers, services, repositories, validators, and DTO mapping, which improved code maintainability and simplified future enhancements

- Optimized SQL Server performance by refining query plans, adding indexes, and tuning stored procedures for reporting and transactional use cases, leading to faster query execution and smoother report generation
- Added JWT-based security with role-based authorization, audit logging, and defensive validation, strengthening endpoint protection and ensuring compliance with data-privacy requirements
- Configured Azure AppInsights monitoring to capture exceptions, dependency timings, and request metrics, enabling faster diagnosis during QA and production verification.
- Worked in Agile sprints, participating in grooming, daily standups, code reviews, and cross functional coordination with QA and product to deliver sprint goals.

GRT Global Logistics

Mar 2022 - Jun 2024

Software Developer

Mumbai

- Built logistics web modules with Angular and .NET Core APIs in Visual Studio, enabling booking, tracking, and workflow automation across shipment lifecycle stages, which streamlined user operations
- Designed service oriented APIs for partner integrations, shipment updates, and warehouse routing, ensuring consistent request models and predictable error handling.
- Implemented Redis caching for frequently accessed operational data and improved API response times by 85% for key read heavy endpoints used across the application.
- Developed message based processing using RabbitMQ and Kafka to handle shipment updates and long running tasks without blocking user requests.
- Designed and maintained SQL Server schemas, stored procedures, indexes, and reporting queries for operational dashboards and business reports, which improved dashboard load times and supported timely decision-making
- Delivered Angular features using reactive forms, nested tables, modular routing while improving perceived responsiveness through lazy loading and request batching.
- Improved UI rendering performance by reducing unnecessary change detection triggers and minimizing redundant HTTP calls through better state handling and caching.
- Owned production support for assigned modules, troubleshoot issues from detection to resolution, and implemented fixes using NUnit tests and Azure DevOps pipelines, reducing module downtime and improving overall system stability

Vinculum Solutions Pvt. Ltd.

Jan 2021 - Mar 2022

Graduate Engineer (Software)

Delhi

- Developed and maintained ASP.NET MVC and Web API modules in C# for loyalty, retail, and commerce products, delivering functional features to multiple customers and enabling seamless transaction processing
- Implemented loyalty capabilities points accrual, redemptions, promotions, and checkout workflows in C# and RxJS, ensuring data accuracy and handling edge cases, which reduced calculation errors for customers
- Integrated payment providers and SMS/email gateways using RabbitMQ and logging pipelines, implemented retry logic and validation, and returned structured error responses, which improved system reliability and reduced integration failures
- Improved database performance by rewriting slow queries, adding appropriate indexes, and applying caching patterns for repeated lookups.
- Supported automated build and deployment workflows with Jenkins and Git, performed release checks and environment validation, which ensured on time releases and reduced deployment errors

Education

Southeast Missouri State University

Aug 2023 - May 2025

M.S., Applied Computer Science

Lovely Professional University

Aug 2017 - May 2021

B.Tech, Computer Science & Engineering

Projects

Online Code Execution Platform (.NET + Angular)

- Built an online code execution portal using ASP.NET Core APIs, Angular, and Docker isolated runtimes to compile and run code securely.
- Integrated Azure SQL for persistence, Blob Storage for artifacts, and Azure Functions for background tasks such as job processing and cleanup.
- Designed API endpoints for submissions, execution results, and user history with validation and structured responses to support a smooth UI experience.